Dissertation Based on Industrial Problem
"Interdependency Analysis of Predominant Meteorological Variables Affecting Rainfall: A Case Study of Surat, Amreli and Gandhinagar"

A dissertation submitted for partial fulfilment of the requirement for the degree Master of Science in Applied Mathematics

by

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CERTIFICATE

This is to certify that the dissertation based on industrial problem entitled "**Interdependency Analysis of predominant meteorological variables affecting rainfall: A case study of Surat, Amreli and Gandhinagar"** which is submitted by **Ms. Chaturvedi Smriti** and **Ms. Borse Sapna** in partial fulfilment forthe award of the degree of **Master of Science** in **Applied Mathematics (Industrial Mathematics)** at **The Maharaja Sayajirao University Of Baroda** has been carried out under my supervision and guidance in the academic year 2023-2024.

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It gives us an immense pleasure and deep satisfaction in presenting **Project on Industrial Problem** entitled **"Interdependency Analysis of predominant meteorological variables affecting rainfall: A case study of Surat, Amreli and Gandhinagar".**

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CHAPTER 1: PROBLEM DESCRIPTION

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1.1 NEED OF STUDY

Rainfall is a significant meteorological parameter, which has direct application in the fields like agriculture and other related divisions. These parts are exceptionally reliant on the accessibility of water just as a satisfactory climatic condition. The measure of precipitation in a region satisfies different needs including farming, industry and pressure driven power age.
Hence, estimating the measure of rainfall plays an important role in the area of agriculture, and industry. Analysis of rainfall data is important as it facilitates decisions regarding cropping patterns, sowing data, construction of roads and providing drinking water to rural and urban areas.

1.2 HISTORICAL BACKGROUND AND LITERATURE REVIEW

Many researches related to rainfall has been done, In the areas of rainfall patterns and trends, impact of climate change, hydrology and water resources, extreme rainfall events, rainfall and public health and many more. Several machine learning approaches were proposed in relation to rainfall prediction. Many studies include rainfall prediction using Artificial Neural Network (ANN).

Pritpal Singh, Bhogeswar Borah (2013)
Forecasting the monsoon temporally is a major scientific issue in the field of monsoon meteorology. The ensemble of statistics and mathematics has increased the accuracy of forecasting of Indian summer monsoon rainfall (ISMR) up to some extent. In this study, we have used feed-forward back-propagation neural network algorithm for ISMR forecasting. Based on this algorithm, we have proposed the five neural network architectures designated as BP1, BP2, BP5 using three layers of neurons (one input layer, one hidden layer and one output layer).

Namitha, K., Jayapriya, A. and Kumar, G.S (2015)
Big data is a celebrated topic in Business as well as research community for several years. With the revolution of Big Data, it is becoming easy and less expensive to store tremendous amount of data for future analysis. Weather data gets accumulated very fast and on a large scale. Thorough analysis and research is required on handling this big data and utilizing it for accurate weather prediction. As deterministic weather forecasting models are usually time consuming,

it becomes challenging to efficiently use this large volume of data in hand. Machine learning methods are already proved to be good replacement for traditional deterministic approaches in weather prediction. These algorithms are popular for their scalability and hence more suitable in big data solutions. This paper proposes an approach of processing such Big volume of weather Data using Hadoop. Proposal includes Artificial Neural Network implemented on Map-reduce framework for short term rainfall prediction. Rainfall is predicted one day ahead using temperature and rainfall data of immediately preceding days. Temperature and Rainfall data of India over past 63 years (1951-2013) is used for this study.

K. Govinda, Shruthi Hiremath (2014)

Rainfall prediction is important because they are used to avoid natural disaster. Rainfall Prediction is the application of science and technology to predict the rainfall for a future time at a given location. Rainfall forecasts are made by collecting quantitative data about the current state of the atmosphere. A neural network can learn complex mappings from inputs to outputs, based solely on samples and require limited understanding from trainer, who can be guided by heuristics. In this paper, possibility of predicting average rainfall has been analyzed through artificial neural network models. In formulating artificial neural network based predictive models three layered network has been constructed.

Shilpa Manandhar, Soumyabrata Dev, Yee Hui Lee, Yu Song Meng, Stefan Winkler (2019)

In recent years, there has been growing interest in using precipitable water vapor (PWV) derived from global positioning system (GPS) signal delays to predict rainfall. However, the occurrence of rainfall is dependent on a myriad of atmospheric parameters. This paper proposes a systematic approach to analyze various parameters that affect precipitation in the atmosphere. Different ground-based weather features such as Temperature, Relative Humidity, Dew Point, Solar Radiation, PWV along with Seasonal and Diurnal variables are identified, and a detailed feature correlation study is presented. While all features play a significant role in rainfall classification, only a few of them, such as PWV, Solar Radiation, Seasonal, and Diurnal features, stand out for rainfall prediction. Based on these findings, an optimum set of features are used in a data-driven machine learning algorithm for rainfall prediction. The experimental evaluation using a 4-year (2012-2015) database shows a true detection rate of 80.4%, a false

alarm rate of 20.3%, and an overall accuracy of 79.6%. Compared to the existing literature, our method significantly reduces the false alarm rates.

Chalachew Muluken Liyew & Haileyesus Amsaya Melese (2021)

Predicting the amount of daily rainfall improves agricultural productivity and secures food and water supply to keep citizens healthy. To predict rainfall, several types of research have been conducted using data mining and machine learning techniques of different countries' environmental datasets. An erratic rainfall distribution in the country affects the agriculture on which the economy of the country depends on. Wise use of rainfall water should be planned and practiced in the country to minimize the problem of the drought and flood occurred in the country. The main objective of this study is to identify the relevant atmospheric features that cause rainfall and predict the intensity of daily rainfall using machine learning techniques. The Pearson correlation technique was used to select relevant environmental variables which were used as an input for the machine learning model. The dataset was collected from the local meteorological office at Bahir Dar City, Ethiopia to measure the performance of three machine learning techniques (Multivariate Linear Regression, Random Forest, and Extreme Gradient Boost). Root mean squared error and Mean absolute Error methods were used to measure the performance of the machine learning model. The result of the study revealed that the Extreme Gradient Boosting machine learning algorithm performed better than others.

C K Gomathy (2021)

Rainfall prediction is important as heavy rainfall can lead to many disasters. The prediction helps people to take preventive measures and moreover the prediction should be accurate. There are two types of prediction short term rainfall prediction and long term rainfall. Prediction mostly short term prediction can gives us the accurate result. The main challenge is to build a model for long term rainfall prediction. Heavy precipitation prediction could be a major drawback for earth science department because it is closely associated with the economy and lifetime of human. It's a cause for natural disasters like flood and drought that square measure encountered by individuals across the world each year. Accuracy of rainfall statement has nice importance for countries like India whose economy is basically dependent on agriculture. The dynamic nature of atmosphere, applied mathematics techniques fail to provide sensible accuracy for precipitation statement. The prediction of precipitation using machine learning techniques may use regression. Intention of this project is to offer non-experts easy

access to the techniques, approaches utilized in the sector of precipitation prediction and provide a comparative study among the various machine learning techniques.

Saranagata Kundu, Saroj Kr. Biswas, Deeksha Tripathi, Rahul Karmakar, Sounak Majumdar , Sudipta Mandal (2023)

Significant challenges to human health and life have arisen as a result of heavy rains. Floods and other natural disasters that affect people all over the world every year are caused by prolonged periods of heavy rainfall. Predictions of rainfall must be accurate in countries like India where agriculture is the primary occupation. The non-linearity of rainfall makes machine learning (ML) methods more efficient than many other approaches. In machine learning (ML), individual classifiers are less accurate than ensemble learning (EL) techniques. In order to better understand the various Machine Learning algorithms and Ensemble Learning techniques that researchers employ to predict rainfall, this review paper has been written. This article reviews ensemble learning algorithms for predicting rainfall. In order to increase the accuracy of rainfall forecasts and consequently avoid the negative effects of heavy precipitation, ensemble learning algorithms have gained popularity. This review article examines and makes reference to the development of ensemble approaches, including bagging, boosting, and stacking. The findings of this survey demonstrate that ensemble techniques are much superior to conventional (individual) model learning in terms of rainfall prediction. Additionally, boosting techniques (such enabling, AdaBoost, and extreme gradient boosting) have been applied more frequently and successfully in scenarios involving rainfall forecasting.

Ari Yair Barrera-Animas, Lukumon O. Oyedele, Muhammad Bilal, Taofeek Dolapo Akinosho, Juan Manuel Davila Delgado, Lukman Adewale Akanbi (2022)

Rainfall forecasting has gained utmost research relevance in recent times due to its complexities and persistent applications such as flood forecasting and monitoring of pollutant concentration levels, among others. Existing models use complex statistical models that are often too costly, both computationally and budgetary, or are not applied to downstream applications. Therefore, approaches that use Machine Learning algorithms in conjunction with time-series data are being explored as an alternative to overcome these drawbacks. To this end, this study presents a comparative analysis using simplified rainfall estimation models based on conventional Machine Learning algorithms and Deep Learning architectures that are efficient

for these downstream applications. Models based on LSTM, Stacked-LSTM, Bidirectional-LSTM Networks, XGBoost, and an ensemble of Gradient Boosting Regressor, Linear Support Vector Regression, and an Extra-trees Regressor were compared in the task of forecasting hourly rainfall volumes using time-series data. Climate data from 2000 to 2020 from five major cities in the United Kingdom were used. The evaluation metrics of Loss, Root Mean Squared Error, Mean Absolute Error, and Root Mean Squared Logarithmic Error were used to evaluate the models' performance. Results show that a Bidirectional-LSTM Network can be used as a rainfall forecast model with comparable performance to Stacked-LSTM Networks. Among all the models tested, the Stacked-LSTM Network with two hidden layers and the Bidirectional-LSTM Network performed best. This suggests that models based on LSTM-Networks with fewer hidden layers perform better for this approach; denoting its ability to be applied as an approach for budget-wise rainfall forecast applications.

M Gowtham Sethupathi, YS Ganesh, MM Ali (2021)

Rainfall prediction is a beneficiary one, but it is a challenging task. Machine learning techniques can use computational methods and predict rainfall by retrieving and integrating the hidden knowledge from the linear and non-linear patterns of past weather data. Various tools and methods for predicting rain are currently available, but there is still a shortage of accurate results. Existing methods are failing whenever massive datasets are used for rainfall prediction. This study provides efficient rainfall prediction methods of machine learning techniques: random forest and logistic regression methods that provide an easy and accurate prediction and determine which one is more effective in comparison. This study would assist researchers in analyzing the most recent work on rainfall prediction with an emphasis on machine learning techniques and providing a reference for possible guidance and comparisons. Anaconda framework is used, and the coding language used is Python, which is portable and dynamic. Numpy, matplotlib, seaborn, and pandas are the libraries used for the implementation.

1.3 PRECIPITATION AND ITS TYPES

Precipitation is liquid or solid water falling from clouds to the earth's surface or formed on different bodies as a result of atmospheric water vapor condensation. Precipitation can be liquid, solid, or mixed. Liquid precipitation includes rain and drizzle (4).

Following are the main types of the precipitation:

- Snow- Ice crystals or flakes that fall from clouds and accumulate on the ground.
- Rainfall- Liquid water droplets that fall from clouds and reach the ground.
- Glaze- A thin layer of ice that forms on surfaces when rain or freezing rain freezes upon contact.
- Sleet- Frozen raindrops or ice pellets that partially melt before reaching the ground.
- Hail- Solid balls or lumps of formed within strong thunderstorm updrafts.

1.4 RAINFALL AND ITS IMPORTANCE

Rainfall is the fall of atmospheric moisture in the form of water due to gravity. Rainfall is the amount of rain that falls in a location over a period of time and therefore a type of precipitation that occurs when water vapor in the atmosphere condenses into droplets that can no longer be suspended in the air. Rainfall is formed when saturated air is heated (air that cools down at dew point) and rises either by a mountain, conventional currents or frontal action. The rising saturated air or water vapor cools down as it rises. It attaches itself to tiny particles of dust, salts, seeds or smoke in the atmosphere. These particles are commonly called condensation nuclei. Condensation takes place when the water droplets join together on the condensation nuclei to form raindrops. Clouds are formed as the rain drops develop which leads to rainfall. The amount of rainfall recorded at a place is measured by an instrument called rain gauge. Rainwater fills reservoirs that supply drinking water, provide a habitat for fish to live, and nourishes the soil with water necessary for vegetation. It is also useful for growing crops and also used for animals and human consumption. It is believed that from all sources of water rain water is the best for use in our homes.

1.5 TYPES OF RAINFALL

There are 3 types of rainfall (4):

- Convective rainfall
- Orographic /Relief rainfall
- Cyclonic /Frontal rainfall

Convective rainfall: Convective rainfall forms convection currents (convection currents describe the rising, spread, and sinking of gas, liquid, or molten material caused by heat application). It occurs when the earth surface becomes more heated than its surrounding. This type of rainfall occurs in equatorial regions and does not last long. This rainfall type occurs in the summer or during the warmer part of the day.

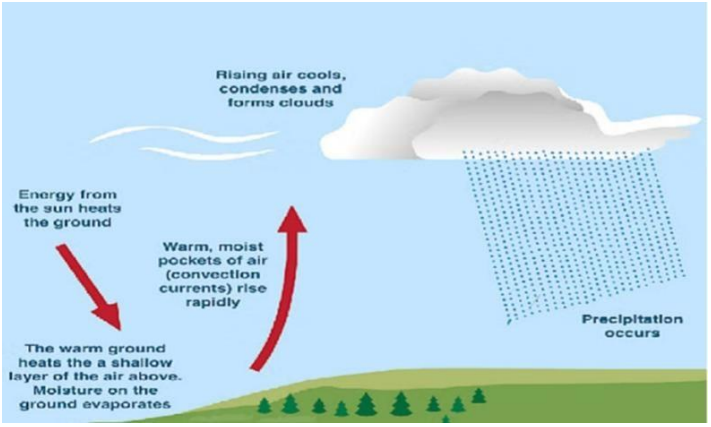

Figure 1: Convective Rainfall

Stages of Convective rainfall:

- The heat of the sun makes the surface of the earth hot. Due to this, the warm air rises.
- As the warm air rises, it starts to cool down. The water vapor then condenses to form clouds.
- There is a certain condensation point to be reached. Here clouds are formed.
- Heavy rainfall starts which includes lightning and thunderstorms. Lightning occurs due to an electrical charge between the clouds.

Orographic/relief rainfall: This type of rainfall occurs when moist air is lifted near a mountain range. It occurs in mountain areas and orographic clouds are formed due to this. It occurs in mountainous areas and along slopes of hill. As the wind or water vapor rises, they become unstable and heavy. They develop around condensational nuclei and form thick clouds. They rise further and become unstable water droplets and fall to ground as raindrops. Humid atmospheric condition and mountain inclination are responsible for this type of rainfall.

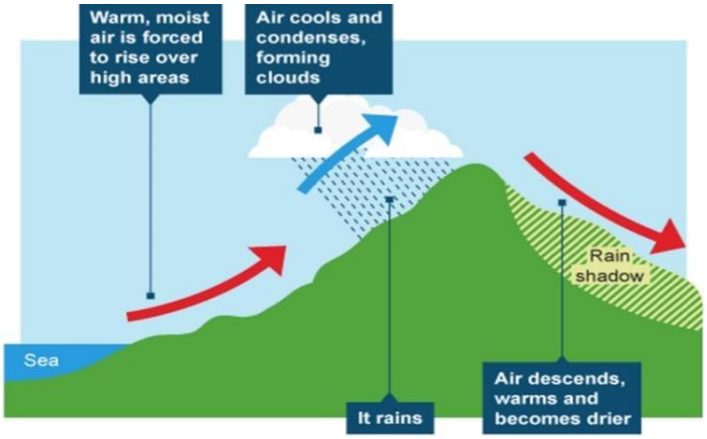

Figure 2: Orographic Rainfall

Stages of orographic rainfall:

- The warm air rises due to the height of the mountains.
- It rises to a certain level and cools down. Due to these clouds are formed.
- The warm air descends.
- As warm air descends, there is little rainfall on one side of the mountain.

Cyclonic/frontal rainfall: cyclonic rainfall is the last type of rainfall and it is also known as frontal rainfall. This type of rainfall occurs when air masses of different characteristics meet i.e. when warm air mass meets cold air mass. Frontal rainfall occurs due to cyclonic activity (A cyclone is a large air mass that rotates along a strong centre of atmospheric pressure. It moves clockwise in the southern hemisphere and anticlockwise in the northern hemisphere). It occurs along the front of cyclone.

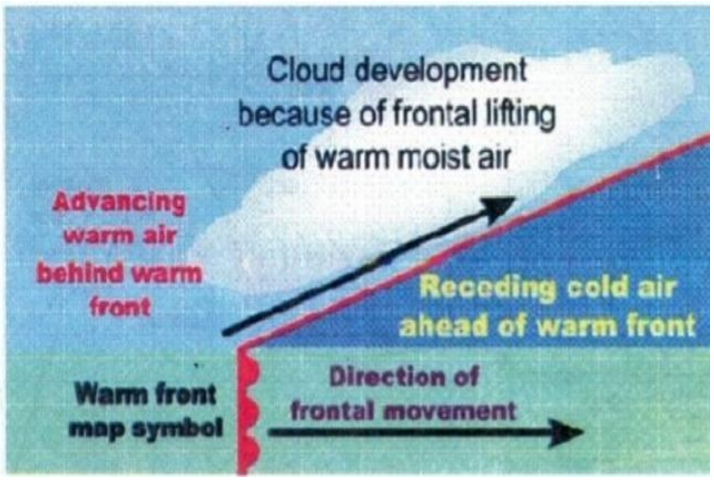

Figure 3: Cyclonic Rainfall

Stages of cyclonic rainfall:

- Warm air meets cool air.
- Warm air rises above the cool air.
- Warm air is cooled and it condenses.
- Clouds start forming and rainfall occurs.

1.6 PARAMETERS CONSIDERED

1.6.1 AVERAGE TEMPERATURE(AT)

The average temperature of the air is indicated by a properly exposed temperature during a given period, usually a day, a month, or a year. To estimate the global temperature, researchers measure air temperatures on land and at sea at thousands of observation stations around the world, day in and day out. Then, they add all those measurements up and divide the result by the number of observations taken that gives an average air temperature for the whole globe. As the average temperature at the earth's surface rises, more evaporation occurs, which in turn increases overall precipitation. Unit: Celsius, Kelvin, Fahrenheit.

Figure 4: Instrument to measure Average Temperature

1.6.2 CLOUD COVER(CC)

Cloud cover refers to the fraction of sky obscured by clouds on average when observed from a particular location (1). Cloud cover is measured using various methods and instruments:

<u>Satellite imagery</u>: -Weather satellites equipped with visible and infrared sensors capture images of the Earth's surface and the atmosphere. By analysing these images, meteorologists can determine the extent of cloud cover.

Figure 5: Satellite Imagery

Sky observation: -Human observers visually estimate cloud cover by dividing the sky into octants and noting the fraction of each segment covered by clouds. This method is often used for weather reports. Instruments for measuring cloud cover are as follow:

Ceilometers: -Ceilometers are ground-based instruments that emit laser or lidar pulses vertically into the atmosphere. They measure the distance to the base of clouds and can provide data on cloud height and cover.

Figure 6: Ceilometer

Radiosondes: - (An instrument carried by ballon or other means to various levels of the atmosphere and transmitting measurements by radio). Weather balloons equipped with radiosondes carry instruments that measure temperature, humidity, and pressure as they ascend through the atmosphere. These measurements can help meteorologists deduce cloud cover and characteristics at different altitudes.

Figure 7: Radiosonde

Automated weather station: -Some automated weather stations have sky cameras that continuously capture images of the sky. Image analysis software can then determine cloud cover by identifying and tracking clouds in the images.

Figure 8: Automated Weather Station

Weather radar: -Weather radar systems can detect precipitation which indirectly indicates cloud cover. The absence of radar echoes in a region suggests a clear sky, while echoes can indicate the pressure of clouds and precipitation.

Figure 9: Weather Radar

The extent of cloud cover determines the amount of moisture in the atmosphere available for condensation. More cloud cover indicates a higher likelihood of condensation and cloud formation. The thickness and density of clouds can influence the intensity of rainfall. Thicker clouds typically contain more moisture and can produce heavier rainfall when they release their moisture. Lighter cloud cover may result in lighter, more scattered showers.

Cloud cover can also influence the type of precipitation. For instance, thicker clouds at colder temperatures are more likely to produce snow, while lighter cloud cover may result in rain or drizzle. Unit of measuring cloud cover is Oktas.

1.6.3 DIURNAL TEMPERATURE RANGE(DTR)

Diurnal range of temperature is the difference between the maximum and minimum temperatures of the same days (6). Diurnal temperature range is measured by the following methods: Diurnal temperature range is measured by recording the daily maximum (highest) temperature and the daily minimum (lowest) temperature over a 24-hour period.

Temperature Sensors: Weather stations and meteorological instruments are equipped with temperature sensors such as thermometers or electronic temperature sensors. These sensors are designed to accurately measure temperature.

Data Collection: Temperature data is collected at regular intervals, usually on an hourly basis, throughout the day and night. These measurements are typically taken in a shaded area away from direct sunlight or other heat sources to ensure accuracy.

Modern weather stations and meteorological agencies use automated systems to continuously monitor and record temperature data, making it easier to calculate diurnal temperature ranges accurately.

Affect of diurnal temperature range on precipitation:

Convection: A larger diurnal temperature range often leads to more intense daytime heating and nighttime cooling. This temperature contrast can enhance atmospheric convection, leading to the development of thunderstorms and precipitation, especially during the warm season.

Orographic Effects: Diurnal temperature variations can impact the movement of air masses. During the day, warm air rises and can interact with topographic features like mountains, causing orographic lifting and increased precipitation on windward slopes.

Moisture Content: Diurnal temperature changes can affect the capacity of the air to hold moisture. Warmer air can hold more moisture, potentially leading to increased humidity and the formation of clouds and precipitation. The specific effects of diurnal temperature range on precipitation can vary depending on local geography, climate patterns, and other factors.

1.6.4 GROUND FROST FREQUENCY(GFF)

Ground frost frequency refers to the number of the ground or surface temperature drops low enough to freeze or form frost (5). It provides information about how frequently the ground, such as soil, grass, or other surfaces, experiences cold enough temperatures to freeze or form frost. Methods of measuring ground frost frequency: Ground frost frequency is calculated as the number of frost events divided by the total number of days or nights in the chosen period. It is often expressed as a percentage or as the number of frost days per year. For example, if there were 30 frost events (nights with temperatures at or below freezing) in a year with 366 days, the ground frost frequency would be calculated as (30/366) x 100%, resulting in a frequency of approximately 8.2%.Ground frost frequency itself does not directly affect rainfall; instead, it is more related to temperature conditions. However, ground frost frequency can indirectly influence certain aspects of the local climate, which may have implications for rainfall patterns. Influence of ground frost frequency on rainfall is complex and can vary depending on regional climate, geography, and other factors. Unit of measuring GFF is days.

1.6.5 MAXIMUM TEMPERATURE(Tmax)

The maximum temperature is the highest recorded or forecasted temperature in a specific location or region over a given period. A common instrument used for measuring temperature is a mercury or alcohol thermometer. In addition to traditional thermometers, modern weather stations often use digital temperature sensors that can automatically record and transmit temperature data to a central database or weather monitoring system. Higher maximum temperatures can lead to increased evaporation rates. Warm air can hold more water vapor than cold air. This can delay the condensation and cloud formation processes that are necessary for precipitation to occur. High temperatures can lead to localized convection, where warm air rises rapidly and creates unstable conditions. This can result in the development of thunderstorms and intense precipitation, especially in the presence of moisture.

1.6.6 MINIMUM TEMPERATURE(Tmin)

The minimum temperature is the lowest recorded or forecasted temperature in a specific location or region over a given period. A common instrument used for measuring temperature is a mercury or alcohol thermometer. In addition to traditional thermometers, modern weather stations often use digital temperature sensors that can automatically record and transmit temperature data to a central database or weather monitoring system. Minimum temperatures can affect factors like snowpack, snowmelt timing, and the overall hydrological cycle, which, in turn, can impact local and regional precipitation patterns.

1.6.7 POTENTIAL EVAPOTRANSPIRATION(PET)

Potential evapotranspiration, or PET, represents the combined loss of water through: 1) the plant's process of transpiration via its vascular system, and 2) evaporation of water from the earth's surface. Both are influenced by temperature, humidity, sunlight, and wind.

Ways of measuring PET:
Meteorological Data: To calculate PET, you need a set of meteorological data, including:
Air temperature

Relative humidity

Wind speed

Solar radiation (or sunshine duration)

Atmospheric pressure

Calculation: The Penman-Monteith equation calculates potential evapotranspiration by taking into account the energy available for evaporation and the resistance to that evaporation. It considers both the energy balance (solar radiation, temperature) and the aerodynamic resistance (wind speed, humidity) of the environment.

Unit: PET is typically expressed in millimetres per day (mm/day) or millimetres per month (mm/month), representing the depth of water that would theoretically evaporate or transpire from a well-watered surface remote sensing data, such as satellite imagery, can be used to estimate PET by providing input data for the Penman-Monteith equation, especially in regions with limited ground-based weather stations. PET is influenced by factors such as temperature, humidity, wind, and solar radiation. High PET values, it indicates that the atmosphere has the capacity to absorb more moisture through evaporation and transpiration. High PET values often coincide with warm and dry conditions, which may be associated with high-pressure systems that can suppress the development of precipitation. reduced vegetation due to prolonged high PET conditions can lead to less transpiration (the release of water vapor by plants) and potentially less moisture being returned to the atmosphere. This can contribute to a local reduction in available moisture for precipitation.

1.6.8 REFERENCE CROP EVAPOTRANSPIRATION(RCET)

The evapotranspiration rate from a reference surface, not short of water, is called the reference crop evapotranspiration or reference evapotranspiration and is denoted as ETo (2). Reference Crop Evapotranspiration (RCET) is not directly measured in the field like some other meteorological parameters; instead, it is calculated using standardized mathematical models and meteorological data. The most commonly used method for calculating RCET is the Penman-Monteith method. The calculated RCET values represent the potential evapotranspiration from the reference crop under standard conditions. It is a parameter used in agricultural and hydrological modelling to estimate the water requirements of crops and the amount of water that needs to be supplied through irrigation**.** RCET is related to crop water management and irrigation, but it doesn't influence rainfall.

1.6.9 VAPOR PRESSURE(VP)

Vapor pressure, is the pressure exerted by a vapor (gaseous phase) when it is in equilibrium with its condensed phase (usually a liquid or solid) in a closed system at a specific temperature.

Methods and instruments of measuring vapor pressure:-

Manometer: A manometer is a simple device used to measure the pressure of gases. To measure vapor pressure, a sample of the substance (liquid or solid) is placed in a closed container, and the vapor pressure is determined by measuring the height of the liquid in a manometer connected to the container. The height of the liquid in the manometer corresponds to the pressure of the vapor above the substance.

Figure 10: Manometer

Knudsen Effusion: This method involves a small hole or orifice in a container separating the vapor phase from the condensed phase. The rate at which molecules escape through this small opening is related to the vapor pressure. By measuring the rate of effusion, vapor pressure can be calculated.

Antoine Equation: For some substances, especially volatile liquids, the Antoine equation is used. It's an empirical equation that relates vapor pressure to temperature. The equation is specific to each substance and is determined through experimental data.

Dynamic Methods: These methods involve continuously measuring the pressure in a closed system while the substance is heated or exposed to changing temperatures. The vapor pressure is then determined by analysing the pressure changes as a function of temperature.

SiteScope: This method involves a special apparatus that measures the temperature and pressure at which a liquid and its vapor are in equilibrium. This data can be used to determine the vapor pressure.

Differential Scanning Calorimetry (DSC): DSC can be used to measure the vapor pressure of a substance by observing the phase transitions (e.g., vaporization) while varying the temperature and pressure within a controlled environment

when air with high vapor pressure comes into contact with a surface with lower vapor pressure (like a body of water), evaporation increases. When the vapor pressure in the atmosphere is high, it means there is more moisture available in the air. High humidity levels are conducive to the formation of clouds because the air can hold more water vapor . Vapor pressure can also affect the intensity of rainfall. Areas with higher vapor pressure are more likely to experience heavy rainfall because there is a greater supply of moisture in the air that can be condensed into raindrops. Areas with consistently high vapor pressure, such as tropical regions, tend to have more frequent and intense precipitation. In contrast, regions with lower vapor pressure may experience less precipitation.

1.6.10 WET DAY FREQUENCY(WDF)

The term “wet day frequency “typically refers to the number of days within a specific period (such as a month or a year) When measurable precipitation, like rain or snow, occurs (3).
Wet day frequency is typically measured using weather data collected from stations or remote sensing instruments.
Data collection: -Weather stations or instruments collect data on precipitation, including the amount of precipitation (Rainfall or snowfall) and the time it occurs. This data is usually Recorded daily.
The threshold definition determines wet days, a precipitation threshold is defined. This threshold represents the minimum amount of precipitation required for a day to be considered “wet”. Common thresholds Include 0.01 inches (0.26 millimetres) or 0.1 millimetres.
Data analysis: -The recorded precipitation data is analysed day by day. If the precipitation amount for a particular day exceeds the defined threshold, that day is counted as a wet day.
Frequency calculation: - The wet day frequency is calculated by counting the number of days that meet the threshold for a specific period, such as month, season, or year.

For example, if you want to calculate the wet day frequency for a specific month, you would count the number of days in that month with precipitation exceeding the defined threshold.

Higher wet day frequency generally means that there are more days with measurable precipitation. This can lead to higher total precipitation. Wet day frequency affects the distribution of rainfall throughout a region. Areas with a high wet day frequency are more likely to experience frequent, but often lighter rainfall events. Conversely, areas with a lower wet day frequency may have in frequent but intense rainfall events. High wet day frequency can increase the likelihood of surface runoff during rain events potentially leading to localized flooding.

1.7 PROBLEM STATEMENT

The objective for Surat, Amreli and Gandhinagar is:

(1)To find the intercorrelation or interdependency between the following parameters.

- Average Temperature(AT)
- Cloud Cover(CC)
- Diurnal Temperature Range(DTR)
- Ground Frost Frequency(GFF)
- Maximum Temperature(Tmax)
- Minimum Temperature(Tmin)
- Potential Evapotranspiration(PET)
- Reference Crop Evapotranspiration(RCET)
- Vapor Pressure(VP)
- Wet Day Frequency(WDF)

(2) To study the above parameters affecting the rainfall in order to find the predominant parameter that causes the rainfall.

1.8 STUDY AREA

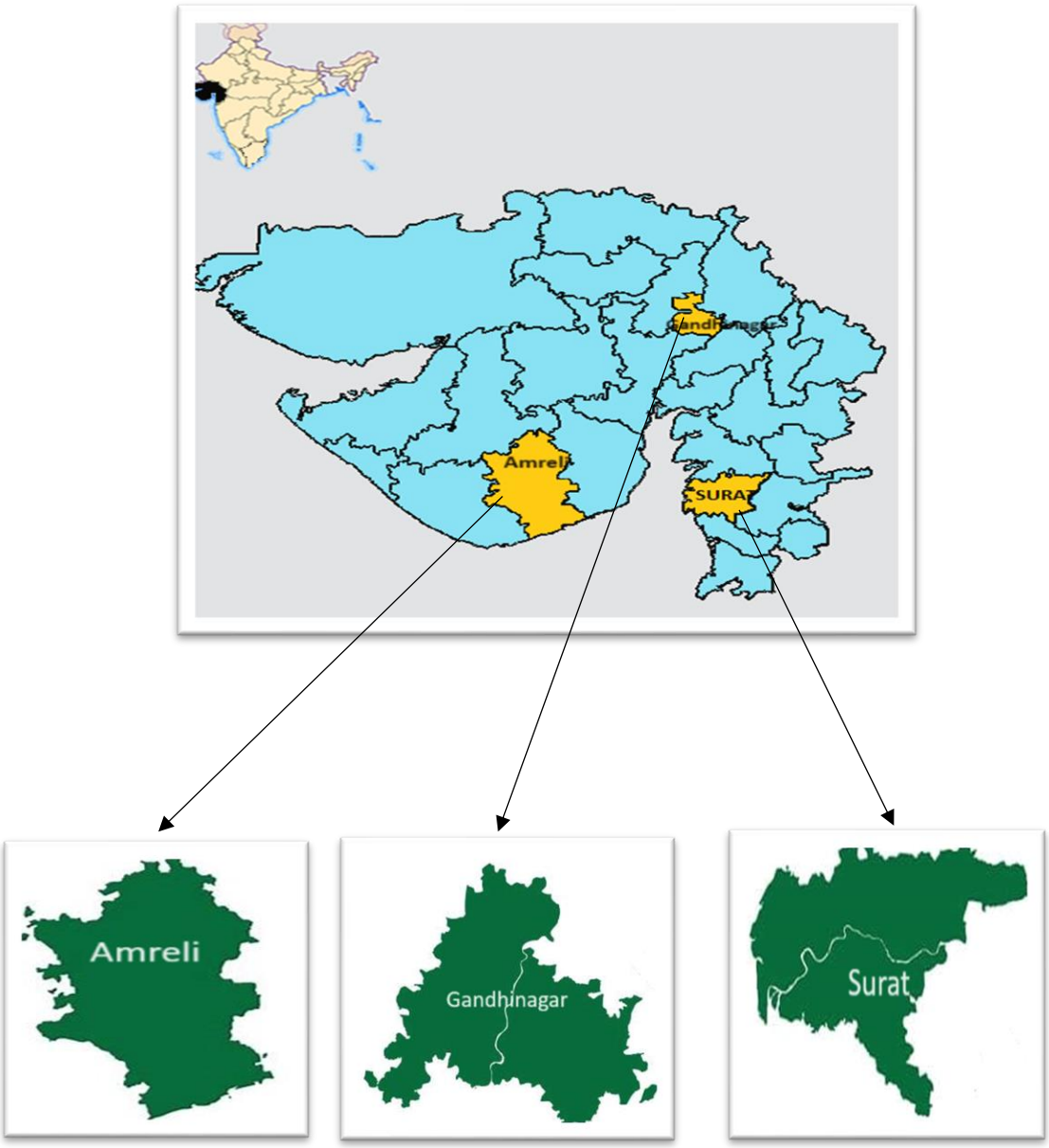

Figure 11: Study Area

Surat is a city located on the western part of India in the state of Gujarat. It is one of the most dynamic city of India with one of the fastest growth rates due to immigration from various part of Gujarat and other states of India. Surat is one of the cleanest city of India and is also known by several other names like “THE SILK CITY”, “THE DIAMOND CITY”, “THE GREEN CITY”, etc. It has the most vibrant present and an equally varied heritage of the past. It is the

city where the British first land in India. The Dutch and the Portuguese also established their business centres in Surat, the remnants of which are still preserved in the modern-day Surat. In past this was a glorious port with ships of more than 84 countries anchored in its harbour at any time. Still today, Surat continues the same tradition as people from all around the country flock in for business and jobs. Surat has practically zero percent unemployment rate and jobs are easier to get here due to very fast development of various industries in and around Surat City. Surat is a famous river port and a settlement with quite an impressive area of over 126 square miles. Latitude and longitude coordinates are: **21.170240, 72.831062**. Located at an elevation of None meters above sea level, Surat has a Tropical steppe wet and dry or savanna ('summer' dry season) climate. The district's yearly temperature is 29.78ºC(85.6ºF) and it is 3.81% higher than India's averages. Surat typically receives about 46.45 millimeter (1.83 inches) of precipitation and has 29.82 rainy days (8.17% of the time) annually.

Amreli district is situated in Saurashtra Region of Gujarat State. District is surrounded by Rajkot and Botad District in North, Junagadh and Gir Somnath District in West and Bhavnagar District in East. Arabian sea touches the district boundary in South. A city of Amreli is one of the most beautiful cultural centres and an industrial hub in central Gujarat, India, with the total population exceeding 100,000 people. The city is clearly divided into a few parts, including a commercial area, a historic centre with many beautiful old buildings, an old king's palace, an old tower, etc. Latitude and longitude coordinates are: **21.603176, 71.222084**. Located at an elevation of None meters above sea level, Amreli has a subtropical steppe climate. The district's yearly temperature is 30.46ºC(86.83ºF) and it is 4.49% higher than India's averages. Amreli typically receives about 14.99 millimeters (0.59 inches) of precipitation and has 20.81 rainy days (5.7% of the time) annually.

Gandhinagar is the capital of the Indian state of Gujarat. It was named after the Father of the Nation, Mahatma Gandhi. Gandhinagar is the second planned city in the western Indian state of Gujarat. Gandhinagar is located on the west bank of the Sabarmati River. Gandhinagar district is surrounded by the districts of Sabarkantha and Aravalli to northeast , Kheda to the southeast ,Ahmedabad to the southwest and Mehsana to the northwest. Gandhinagar district is an administrative division of Gujarat, India with the total population exceeding . Latitude and Longitude coordinates are:**23.2156, 72.6369.** Located at an elevation of None meters above sea level, Gandhinagar has a subtropical steppe climate. The city lies on the bank of river. The

district's yearly temperature is 31.17ºC(88.11ºF) and it is 5.2% higher than India's averages. Gandhinagar typically receives about 30.25 millimeter (1.19 inches) of precipitation and has 22.99 rainy days (6.3% of the time) annually.

1.9 SAMPLE DATA AND SOURCE OF DATA

Month	AT (ºC)	CC (okta)	DTR (ºC)	GFF (Days)	Tmax (ºC)	Tmin (ºC)	PET (mm/ unit time)	RCET (mm/ unit time)	VP (N/m^2)	WDF (mm/day)	Precipitation (mm)
Jan	16.204	9.306	12.436	0.3516	22.459	10.021	4.66	2.88	10.343	0.5237	2.714
Feb	16.706	9.101	12.66	0.325	23.035	10.376	5.14	3.41	10.29	0	0
Mar	21.064	14.99	12.037	0.053	27.031	15.047	5.82	4.43	12.906	0.3737	2.115
Apr	23.017	15.459	10.836	0.0734	28.427	17.588	6.26	4.93	16.422	0.5643	0.776
May	24.436	18.931	9.145	0.0212	29.04	19.871	6.14	4.99	19.689	5.0425	1.595
Jun	24.286	40.06	6.865	0	27.756	20.864	5.23	4.38	24.144	11.8743	77.731
Jul	22.351	59.326	5.114	0	24.912	19.799	4.2	3.56	24.789	11.8743	445.8
Aug	20.991	57.937	4.827	0.0105	23.403	18.578	3.87	3.21	22.396	9.1384	194.179
Sep	21.355	42.53	6.475	0.0115	24.598	18.124	4.34	3.45	21.757	2.8602	20.291
Oct	22.048	18.014	9.801	0.0186	26.968	17.167	5.17	3.76	11.878	1.1071	19.766
Nov	19.903	11.222	11.346	0.0718	25.59	14.245	4.93	3.27	13.593	0.1966	0.196
Dec	18.192	10.062	12.031	0.0131	24.229	12.198	4.59	2.88	11.89	0	0

Table 1: Sample Data

Above table represents the sample data of one year of rainfall i.e. 1901 and the parameters affecting the rainfall. The monthly data of 102 years i.e. from 1901 to 2002 of all the parameters considered are collected from the "Online portal of IMD(Indian Meteorological Department) (12) ".

LAYOUT OF THE THESIS

CHAPTER 1: PROBLEM DESCRIPTION

It includes the historical background of the study, Precipitation and its forms. Insight of the importance of rainfall and its types and a brief understanding of the parameters affecting the rainfall and how it is measured. It also includes the problem definition followed by the study area sample data and its source.

CHAPTER 2: MATHEMATICAL PRELIMINARIES & RESEARCH METHODOLOGIES

This chapter includes the fundamental concepts of linear algebra like Matrices, eigen values, eigen vectors, linear dependence etc. as well as the concepts of statistical analysis. Understanding of these concepts have been used to develop a methodology and has been applied to the data of the considered parameters for Surat, Amreli and Gandhinagar.

CHAPTER 3: RESULTS AND DISCUSSION

This chapter includes the results obtained by implementing the methodology to the data of Surat, Amreli and Gandhinagar. Additionally, it includes a thorough discussion and interpretation of the obtained results followed by the conclusion.

CHAPTER 2 : MATHEMATICAL PRELIMINARIES AND RESEARCH METHODOLOGY

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In this chapter we have discussed foundational concepts of mathematics that help us understanding both linear algebra and statistical analysis. These mathematical preliminaries form the essential framework upon which we build our understanding and application of data analysis techniques. These mathematical preliminaries related to linear algebra and statistical analysis have been applied to the data.

2.1 MATHEMATICAL PRELIMINARIES

2.1.1 AN INTRODUCTION TO MATRIX

A matrix is a rectangular array of numbers

$$A=\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}$$

Where the a_{ij} are scalars, called the entries or components of A. The size or dimension of a matrix is defined as m×n, where m is the number of rows and n is the number of columns. When specifying the size of a matrix, we always list the number of rows before the number of columns. The (i, j)-entry of a matrix is the entry in the ith row and j^{th} column, and is often denoted a_{ij} . We sometimes use A=[a_{ij}] as a short-hand notation for the entire m×n matrix whose (i, j)-entry is equal to ai j for all i = 1, . . .m and j = 1, . . . ,n. There are various operations which are done on matrices of appropriate sizes. Matrices can be added and subtracted, multiplied by a scalar, and multiplied by other matrices.

2.1.2 DIAGONAL MATRIX

A square matrix in which all the elements except the principal diagonal elements are zero. i.e. a_{ij}=0, where i≠j.

$$A=\begin{bmatrix} a_{11} & 0 & \cdots & 0 \\ 0 & a_{22} & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & a_{mm} \end{bmatrix}$$

2.1.3 UPPER TRIANGULAR MATRIX AND LOWER TRIANGULAR MATRIX

Upper triangular matrix:

A square matrix in which all the elements below diagonal are zero.
i.e. a_{ij}=0, where i>j or a_{ij} ≠0, where i≤j

$$A=\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ 0 & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & a_{mn} \end{bmatrix}$$

Lower triangular matrix:

A square matrix in which all the elements above diagonal are zero.
i.e. a_{ij}=0, where i<j or a_{ij} ≠0, where i≥j

$$A=\begin{bmatrix} a_{11} & 0 & \cdots & 0 \\ a_{21} & a_{22} & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}$$

2.1.4 THE TRANSPOSE OF A MATRIX:

Let A be an m×n matrix. Then the transpose of A, denoted AT, is the n×m-matrix whose (i,j)-entry is the (j, i)-entry of A.

$$A=\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}$$

and

$$A^T=\begin{bmatrix} a_{11} & a_{21} & \cdots & a_{m1} \\ a_{12} & a_{22} & \dots & a_{m2} \\ \vdots & \vdots & \ddots & \vdots \\ a_{1n} & a_{2n} & \cdots & a_{mn} \end{bmatrix}$$

2.1.5 SYMMETRIC AND SKEW-SYMMETRIC MATRICES

An n×n matrix A is said to be symmetric if A^T = A. It is said to be skew-symmetric if $A^T = -$A.

2.1.6 TRACE OF A MATRIX

Let A be a square matrix of order nxn then its trace is defined as sum of the diagonal elements (19)

$$A=\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix}$$

Trace(A)=a_{11}+a_{22}+… +a_{nn}

2.1.7 RANK OF A MATRIX

Let A be any mxn matrix and let t be any natural number such that
t ≤ min(m,n), where min (m,n) denotes the smaller of m and n.
If now, we delete any (m-t) rows and (n-t) columns of the matrix, then the remaining elements form a t-rowed square submatrix whose determinant is a minor of the matrix A of order t. Thus, to each admissible value of t corresponds a system of minors of A.

Definition of Rank of a matrix :

The number r with the following two properties is called the rank of the matrix:

1. There is at least one minor of order r, which does not vanish.
2. Every minor of order (r+1) vanishes.

This definition of rank does uniquely fix the same for, as a consequence of (2), Every minor of order (r+2), being the sum of multiples of minor of order (r+1), will vanish. In fact, every minor of order greater than r will vanish as a consequence of (2). Briefly we may say that the rank of a matrix is the largest order of any non-vanishing minor of the matrix.

The following two simple results follow immediately from the definition:

The rank is ≥ r, if there exists a non-zero minor of order r.

The rank is ≤ r, if all minors of order (r+1) are zero.

In each of these two cases above we assume, r to satisfy only one of the two properties (1) or (2) of the rank.

Two extreme cases present themselves as follows:

- As the rank of every non-zero matrix is ≥ 1, we agree to assign the rank ,0 to every zero matrix.
- As the rank of a matrix whose minors of order n are all zero is < n, we agree to assign rank n to the matrix having at least one non-zero minor of order n; n being the highest order of any minor of the matrix.
- The rank of every n-rowed non-singular matrix A is n. The rank of matrix A may be denoted by the symbol ϱ(A).

2.1.8 THE INVERSE OF A MATRIX

Let A and B be n×n matrices. We say that B is an inverse of A if

BA = I and AB = I.

If this is the case, we also write B = A^{-1}. When a matrix has an inverse, it is called invertible.

2.1.9 EIGEN VALUES AND EIGEN VECTORS

Let A be an n×n matrix. Suppose that v ∈ R^n is a non-zero vector such that Av is a scalar multiple of v. In other words, suppose that there exists a scalar λ such that Av = λv Then v is called an eigenvector of A, and λ is called the corresponding eigen value.

1. λ is an eigenvalue of A.
2. There exists a non-zero vector v such that Av = λv.
3. The homogeneous system of equations (A−λI)v = 0 has a non-trivial solution

If A is a square matrix, and let λ is a scalar. Then λ is an eigenvalue of A if and only if det(A−λI) = 0.

2.1.10 LINEAR DEPENDENCE AND LINEAR INDEPENDENCE

Linear dependence

Let V(F) be a vector space. A finite set $(x_1 \;\; x_2 \;\; \dots \;\; x_n)$ of vectors of V is said to be linearly dependent, if there exist scalars $a_1, a_2, \dots, a_n \in$ F not all of them 0 (some of them may be zero) such that $a_1x_1 + a_2x_2 + a_3x_3 + \cdots + a_nx_n = 0$

Linear independence:

Let V(F) be a vector space. A finite set $(x_1 \ x_2 \ \ldots \ x_n)$ of vectors of V is said to be linearly independent if every relation of the form

$a_1x_1 + a_2x_2 + a_3x_3 + \cdots + a_nx_n = 0$, $a_i \in$ F , 1≤ i ≤ n

=>$a_i = 0$ for each 1 ≤ i ≤ n.

Any infinite set of vectors of V is said to be linearly independent if its every finite subset is linearly independent, otherwise it is linearly dependent.

2.1.11 DIAGONALIZABLE MATRIX

Let A be an n×n matrix. Then A is said to be diagonalizable if there exists an invertible matrix P and a diagonal matrix D such that P^{-1}AP = D.

An n×n matrix A is diagonalizable if and only if A has n linearly independent eigenvectors.

let P be the invertible matrix whose columns are n linearly independent eigenvectors of A, and let D be the diagonal matrix whose diagonal entries are the corresponding eigenvalues. Then P^{-1}AP = D.

2.1.12 PROCESS OF DIAGONALIZATION

Consider a matrix A of order n×n

$$A=\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \ldots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix}$$

Step 1: Find the eigenvalues of A

Step 2: Find n linearly independent eigenvectors of A i.e. $v_1, v_2, v_3, \ldots, v_n$.

Step 3: Construct a matrix P from eigen vectors $v_1, v_2, v_3, \ldots, v_n$ found in step 2

P= $[v_1, v_2, v_3, \ldots, v_n]$

Step 4: Construct D from the corresponding eigenvalues.

D=P^{-1}AP

2.1.13 VARIANCE

Another way of dealing with the positive and negative deviations is to square them, since both $(+a)^2$ and $(-a)^2$ are equal to a^2. The sum of the squares of the deviation is divided by one less than the number of individuals, and the resulting measure of variability is called the variance (20). In usual notation,

$$\text{Variance}=\frac{\sum_{i=1}^{n}(\mathrm{X}-\bar{X})^2}{n-1}$$

2.1.14 STANDARD DEVIATION

The variance is in squares of these units but can be reduced to the original units by taking the square root. The square root of the variance is called the standard deviation(s), and is the most widely used of the measures of variability.

$$\text{Standard deviation(s)}=\sqrt{\frac{\sum_{i=1}^{n}(\mathrm{X}-\bar{X})^2}{n-1}}$$

2.1.15 COVARIANCE

Covariance is defined as a measure of how much two variables change together i.e. expected value of variations of two variables from their expected value. The mean of each variable is used as reference and relative positions of observations compared to mean is important. Covariance is simply defined as the mean of multiplication of corresponding X and Y deviations from their mean. Covariance is expressed as following formula (20):

$$\text{Covariance(X,Y)} = \frac{\sum_{i=1}^{n}(X-\bar{X})(Y-\bar{Y})}{n-1}$$

Where,

$\bar{X}$= Mean of values in X

$\bar{Y}$= Mean of values in Y

n=Number of pairs (x,y) , x∈X , y∈Y

2.1.16 COVARIANCE MATRIX

A covariance matrix is a fundamental concept in statistics and machine learning, used to understand the relationship between multiple variables. A covariance matrix is a square matrix giving the covariance between each pair of elements. It is a symmetric matrix with its diagonal entries equal to the variance of the element.

X=[$x^{(1)}$, $x^{(2)}$, $x^{(3)}$, …,$x^{(m)}$]

Given a matrix X where each column represents a variable and each row represents an observation, then Covariance matrix corresponding to X is given by:

$$S_x = X^T X = \begin{bmatrix} x^{(1)} \\ \vdots \\ x^{(m)} \end{bmatrix} [x^{(1)} \ \dots \ x^{(m)}]$$

$$S_x = \begin{bmatrix} < x^{(1)}, x^{(1)} > & < x^{(1)}, x^{(2)} > & \cdots & < x^{(1)}, x^{(m)} > \\ < x^{(2)}, x^{(1)} > & < x^{(2)}, x^{(2)} > & \dots & < x^{(2)}, x^{(m)} > \\ \vdots & \vdots & \ddots & \vdots \\ < x^{(m)}, x^{(1)} > & < x^{(m)}, x^{(2)} > & \cdots & < x^{(m)}, x^{(m)} > \end{bmatrix}$$

Properties of covariance matrix:

- The covariance matrix S_x is a symmetric positive semi definite matrix with non-negative eigen value.
- The diagonal entries of the covariance matrix represent the variance of each variable.
- The off-diagonal entries represent the covariance between pairs of variables.

2.1.17 CORRELATION

Correlation is one of the most useful statistical tools which allows us to make comparison between two sets of data.

With the help of correlation, we can determine whether the two different quantities are related to each other. Correlation coefficient measures the degree of relationship or correlation between the two quantities. Correlation coefficient is denoted by r and it has a numerical value from -1 and 1. If the value for r comes out close to 1 , then there is a strong positive correlation between two quantities. On the other hand , If the value for r is close to -1 then there is a strong negative correlation between the two quantities. If the value for r is 0 then there is a strong that two quantities are not related. Positive correlation between two quantities indicates that increase in

one quantity also increases the second quantity similarly decrease in the quantity indicates the decrease in second quantity. That is negative correlation suggests If there is positive correlation then two quantities are proportional to each other while negative correlation suggests that two quantities are inversely proportional. The correlation coefficient r is defined as

$$r = \frac{\sum_{i=1}^{n}(X-\bar{X})(Y-\bar{Y})}{\sqrt{\sum_{i=1}^{n}(X-\bar{X})^2}\sqrt{\sum_{i=1}^{n}(Y-\bar{Y})^2}} \quad , -1 \leq r \leq 1$$

Where , X and Y are the quantities whose correlation is to be found. $\bar{X}$is the mean of X values where as $\bar{Y}$ is the mean of Y values. We can reduce this to

$$r = \frac{n\sum_{i=1}^{n}(XY)-(\sum_{i=1}^{n}X)(\sum_{i=1}^{n}Y)}{\sqrt{n(\sum_{i=1}^{n}X^2)-(\sum_{i=1}^{n}X)^2}\sqrt{n(\sum_{i=1}^{n}Y^2)-(\sum_{i=1}^{n}Y)^2}} = \frac{COV(X,Y)}{\sigma_X\sigma_Y}$$

Here, n represents the number of the data pairs. σ_X and σ_Y represents the standard deviations of X and Y respectively. The values of r=-1 and r=1 occur only in case of a perfect linear relationship. Correlation measures the strength of only a linear relationship between two variables. It does not describe curved relationship, no matter how strong they are. Correlation coefficient r uses the standardized values of the observations. It does not change when we change unit of measurements.

2.1.18 CORRELATION MATRIX

The correlation matrix plays an important role in multivariate analysis. Since by itself it captures the pairwise degrees of relationship between different variables. It is used to test the independence of variables. If correlation value between two variable is zero then it indicates that two variables are independent. Correlation matrix is a symmetric matrix with diagonal entries equal to one as it shows correlation of a variable with itself. Consider n variables, n∈ N(N is the set of natural numbers).

Correlation matrix = $\begin{bmatrix} 1 & a_{12} & \cdots & a_{1n} \\ a_{21} & 1 & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & 1 \end{bmatrix}$

Where a_{12} indicates correlation of first variable with the second variable a_{13} indicates correlation between first and third variable and so on. In general, a_{mn} indicates correlation between variables m and n ,where m≤n.

1.1.19 RELATIVE MEASURE OF VARIATION :

Consider a diagonalized matrix of order m

D = $\begin{bmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_m \end{bmatrix}$

The relative measure of variation is defined as the ratio of diagonal element to the trace of the diagonal matrix i.e. $R_k = \frac{\lambda_k}{tr(D)}$, 1≤ k ≤m.

2.2 RESEARCH METHODOLOGY

The present study has been carried out to find the intercorrelation between the parameters affecting the rainfall and also to find the predominant parameters affecting the rainfall. Data from year 1901 to 2002 for Surat, Amreli and Gandhinagar districts in Gujarat were analysed.

METHODOLOGY FOR OBJECTIVE 1

The study to find the intercorrelation or interdependency between the considered parameters has been undertaken by finding the correlation matrix of the data as per the method given in 2.1.18.

METHODOLOGY FOR OBJECTIVE 2

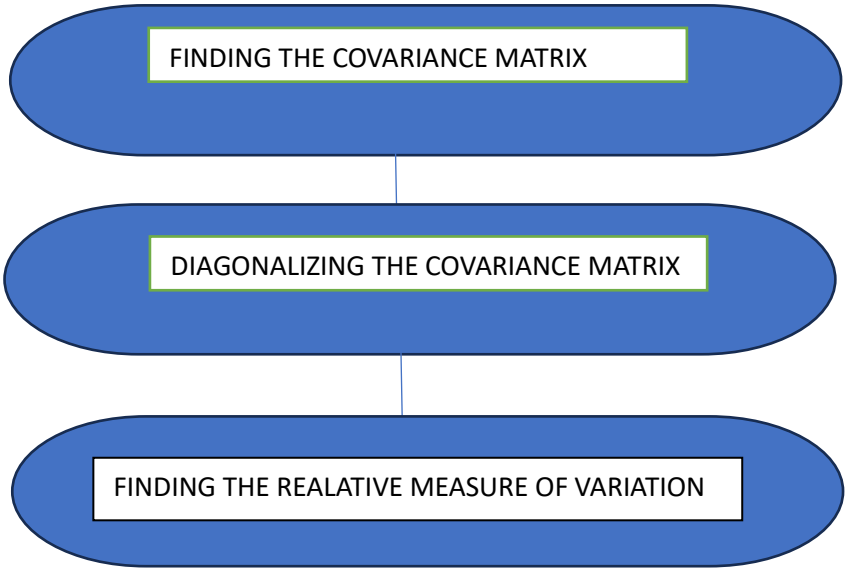
predominant parameters have been found by following the following steps.

1) Calculating the covariance matrix of the data , separately for Surat, Amreli and Gandhinagar in order to know the covariance between all the pairs of parameters.

2) Diagonalizing the covariance matrix to simplify analysis.

3) Determining the relative measure of variation to ascertain the significance of parameters in influencing rainfall.

Flow chart for finding the relative measure of variation



Flowchart 1

2.2.1 METHODOLOGY FOR OBJECTIVE 1

For the data of Surat

	AT	CC	DTR	GFF	Tmax	Tmin	PET	RCET	VP	WDF	Precipitation
AT	1	0.382	-0.523	-0.5	0.785	0.928	0.417	0.791	0.656	0.247	**0.212**
CC	0.382	1	-0.963	-0.398	-0.255	0.687	-0.6	-0.092	0.877	0.929	**0.815**
DTR	-0.523	-0.963	1	0.424	0.118	-0.802	0.511	-0.004	-0.926	-0.882	-0.767
GFF	-0.5	-0.398	0.424	1	-0.273	-0.535	0.101	-0.167	-0.474	-0.339	-0.287
Tmax	0.785	-0.255	0.118	-0.273	1	0.499	0.856	0.918	0.092	-0.353	-0.31
Tmin	0.928	0.687	-0.802	-0.535	0.499	1	0.07	0.556	0.864	0.558	**0.483**
PRT	0.417	-0.6	0.511	0.101	0.856	0.07	1	0.849	-0.279	-0.643	-0.546
RCET	0.791	-0.092	-0.004	-0.167	0.918	0.556	0.849	1	0.234	-0.202	-0.164
VP	**0.656**	**0.877**	-0.926	-0.474	0.092	**0.864**	-0.279	0.234	1	**0.781**	**0.686**
WDF	0.247	**0.929**	-0.882	-0.339	-0.353	0.558	-0.643	-0.202	0.781	1	**0.94**
PRECIPITATION	0.212	0.815	-0.767	-0.287	-0.31	0.483	-0.546	-0.164	0.686	0.94	1

Table 2: Correlation Matrix for the data of Surat

The above table is a correlation matrix found by method stated in 2.1.18, it shows the correlation between the pairs of considered parameters.

Where,

AT= Average Temperature

CC= Cloud Cover

DTR= Diurnal Temperature Range

GFF= Ground Frost Frequency

Tmax= Maximum Temperature

Tmin= Minimum Temperature

PET= Potential Evapotranspiration

RCET= Reference Crop Evapotranspiration

VP= Vapor Pressure

WDF= Wet Day Frequency

It can be clearly seen that the precipitation is positively correlated with Wet Day Frequency, Vapor Pressure, Cloud Cover ,Minimum Temperature and Average Temperature as correlation coefficient of precipitation with Wet Day Frequency, Vapor Pressure, Cloud Cover, Minimum Temperature is 0.94 , 0.686 , 0.815 , 0.483 and 0.212 respectively . However, Wet Day Frequency is strongly correlated with Cloud Cover as correlation coefficient of Wet Day Frequency with Cloud Cover is 0.929 whereas Vapor Pressure is related with Cloud Cover, Minimum Temperature, Wet Day Frequency and Average Temperature as correlation coefficient of Vapor Pressure with Cloud Cover, Minimum Temperature, Wet Day Frequency and Average Temperature is 0.877,0.864,0.781,0.656 respectively.

For the data of Amreli

	AT	CC	DTR	GFF	Tmax	Tmin	PET	RCET	VP	WDF	Precipitation
AT	1	0.592	-0.794	-0.524	0.745	**0.97**	-0.04	0.556	**0.917**	0.497	**0.425**
CC	0.592	1	-0.928	-0.359	-0.064	0.724	-0.748	-0.138	0.79	0.875	**0.7**
DTR	-0.794	-0.928	1	0.408	-0.211	-0.904	0.544	-0.121	-0.949	-0.786	-0.636
GFF	-0.524	-0.359	0.408	1	-0.409	-0.511	0.17	-0.146	-0.451	-0.284	-0.229
Tmax	0.745	-0.064	0.211	-0.409	1	0.609	0.545	0.793	0.473	-0.115	-0.075
Tmin	0.97	0.724	-0.904	-0.511	0.609	1	-0.202	0.446	0.977	0.587	**0.483**
PRT	-0.04	-0.748	0.544	0.17	0.545	-0.202	1	0.756	-0.275	-0.71	-0.547
RCET	0.556	-0.138	-0.121	-0.146	0.793	0.446	0.756	1	0.394	-0.215	-0.146
VP	0.917	0.79	-0.949	-0.451	0.473	0.977	-0.275	0.394	1	0.643	**0.534**
WDF	0.497	**0.875**	-0.786	-0.284	-0.115	0.587	-0.71	-0.215	0.643	1	**0.907**
PRECIPITATION	0.425	0.7	-0.636	-0.229	-0.075	0.483	-0.547	-0.146	0.534	0.907	1

Table 3: Correlation Matrix for the data of Amreli

The above table is a correlation matrix found by method stated in 2.1.18, it shows the correlation between the pairs of parameters for the data of Amreli. For the data of Amreli the precipitation is positively correlated with Wet Day Frequency, Cloud Cover, Vapor Pressure, Minimum Temperature and Average Temperature as correlation coefficient of precipitation with is 0.907 , 0.7 , 0.534 , 0.483 and 0.425 respectively . However, Wet Day Frequency is strongly correlated with Cloud Cover as correlation coefficient of Wet Day Frequency with Cloud Cover is 0.875 . While , Average Temperature is strongly related with Minimum Temperature and Vapor Pressure as correlation coefficient of Average Temperature with Minimum Temperature and Vapor Pressure is 0.97 and 0.917 respectively.

For the data of Gandhinagar

	AT	CC	DTR	GFF	Tmax	Tmin	PET	RCET	VP	WDF	Precipitation
AT	1	0.47	-0.47	-0.47	0.64	**0.96**	0.64	0.88	0.77	0.32	**0.25**
CC	0.47	1	-0.96	-0.96	-0.31	0.69	-0.31	0.13	0.87	0.89	**0.77**
DTR	-0.47	-0.96	1	1.00	0.35	-0.70	0.35	-0.07	-0.90	-0.87	-0.75
GFF	-0.47	-0.96	1.00	1	0.35	-0.70	0.35	-0.07	-0.90	-0.87	-0.75
Tmax	0.64	-0.31	0.35	0.35	1	0.41	1.00	0.90	0.06	-0.38	-0.33
Tmin	0.96	0.69	-0.70	-0.70	0.41	1	0.41	0.73	0.91	0.53	**0.44**
PET	0.64	-0.31	0.35	0.35	1.00	0.41	1	0.90	0.06	-0.38	-0.33
RCET	0.88	0.13	-0.07	-0.07	0.90	0.73	0.90	1	0.46	-0.01	-0.02
VP	**0.77**	**0.87**	-0.90	-0.90	0.06	**0.91**	0.06	0.46	1	**0.74**	**0.63**
WDF	0.32	**0.89**	-0.87	-0.87	-0.38	0.53	-0.38	-0.01	0.74	1	**0.92**
PRECIPITATION	0.25	0.77	-0.75	-0.75	-0.33	0.44	-0.33	-0.02	0.63	0.92	1

Table 4: Correlation Matrix for the data of Gandhinagar

The above table is a correlation matrix found by method stated in 2.1.18, it shows the correlation between the pairs of parameters for the data of Gandhinagar . It can be clearly seen that the precipitation is positively correlated with Wet Day Frequency, Cloud Cover, Vapor Pressure, Minimum Temperature and Average Temperature as correlation coefficient of precipitation with Wet Day Frequency, Cloud Cover, Vapor Pressure, Minimum Temperature and Average Temperature is 0.92, 0.77 , 0.63 , 0.44 and 0.25 respectively . However, Wet Day Frequency is strongly correlated with Cloud Cover as correlation coefficient of Wet Day Frequency with Cloud Cover is 0.89. While Vapor Pressure is strongly related with Minimum Temperature, Cloud Cover, Average Temperature and Wet Day Frequency as correlation coefficient of Vapor Pressure with Minimum Temperature, Cloud Cover, Average Temperature and Wet Day Frequency is 0.91,0.87,0.77,0.74 respectively and Average temperature is strongly related with Minimum Temperature as correlation coefficient of Average Temperature with Minimum Temperature is 0.96.

2.2.2 METHODOLOGY FOR OBJECTIVE 2

For the data of Surat

<u>Finding the relative coefficient of variation</u>

Step 1: Finding the covariance matrix:

A covariance matrix is a square matrix giving the covariance between each pair of elements. It is a symmetric matrix with its diagonal entries equal to the variance of the element. Below table is the covariance matrix for the data of Surat it shows covariance between a pair of parameter. It can be seen from the below table that the value 13.777 shows the covariance of Vapor pressure and Minimum temperature similarly all the nondiagonal entries shows the covariance between two different parameters . The diagonal entries of the below matrix give the variance of the data of the different parameters. Below matrix is a is a symmetric positive semi definite matrix with non-negative eigen value.

	AT (ºC)	CC (okta)	DTR (ºC)	GFF (Days)	Tmax (ºC)	Tmin (ºC)	PET (mm/unit time)	RCET (mm/unit time)	VP (N/m^2)	WDF (mm/day)	Precipitation (mm)
AT	5.062	15.747	-3.293	-0.087	3.414	6.7	0.692	1.238	7.335	2.648	76.046
CC	15.747	336.228	-49.448	-0.567	-9.041	40.432	-8.114	-1.173	79.908	81.167	2380.3
DTR	-3.293	-49.448	7.843	0.092	0.636	-7.207	1.056	-0.009	-12.885	-11.772	-342.138
GFF	-0.087	-0.567	0.092	0.006	-0.041	-0.133	0.006	-0.009	-0.183	-0.126	-3.544
Tmax	3.414	-9.041	0.636	-0.041	3.734	3.091	1.222	1.234	0.887	-3.253	-95.427
Tmin	6.7	40.432	-7.207	-0.133	3.091	10.293	0.165	1.242	13.777	8.526	246.91
PET	0.692	-8.114	1.056	0.006	1.222	0.165	0.545	0.436	-1.023	-2.26	-64.173
RCET	1.238	-1.173	-0.009	-0.009	1.234	1.242	0.436	0.484	0.808	-0.67	-18.123
VP	7.335	79.908	-12.885	-0.183	0.887	**13.777**	-1.023	0.808	24.688	18.494	542.719
WDF	2.648	81.167	-11.772	-0.126	-3.253	8.526	-2.26	-0.67	18.494	22.721	713.441
PRECIPITATION	76.046	2380.3	-342.138	-3.544	-95.427	246.91	-64.173	-18.123	542.719	713.441	25349.2

Table 5: Covariance Matrix for the data of Surat

Step 2: Diagonalizing the covariance matrix

2.0919	0	0	0	0	0	0	0	0	0	0
0	0.0026	0	0	0	0	0	0	0	0	0
0	0	6.9×10^{-5}	0	0	0	0	0	0	0	0
0	0	0	8.1×10^{-5}	0	0	0	0	0	0	0
0	0	0	0	9.6×10^{-4}	0	0	0	0	0	0
0	0	0	0	0	0.0785	0	0	0	0	0
0	0	0	0	0	0	0.4897	0	0	0	0
0	0	0	0	0	0	0	0.7978	0	0	0
0	0	0	0	0	0	0	0	16.01	0	0
0	0	0	0	0	0	0	0	0	127.8	0
0	0	0	0	0	0	0	0	0	0	25613.4

Table 6: Diagonalized Covariance Matrix for the data of Surat

The above Matrix is obtained by diagonalizing the covariance matrix found in step 2by the procedure stated in 2.1.12. Where the diagonal entries represent the eigen values of the covariance matrix.

Step 3: Finding the relative measure of variation

The relative measure of variation is defined as the ratio of diagonal element to the trace of the diagonal matrix i.e. $R_k = \frac{\lambda_k}{tr(D)}$, $1 \leq k \leq m$. The trace of the above diagonal matrix is **25760.79034** hence the relative measure of variation is calculated by:

$R_k = \frac{\lambda_k}{\mathbf{25760.79034}}$. The relative measure of the variation for the above diagonalized matrix is

$R_{AT} = R_1 = \frac{\lambda_1}{\mathbf{25760.79034}} = 8.1028\times 10^{-5}$

$R_{CC} = R_2 = \frac{\lambda_2}{\mathbf{25760.79034}} = 1.0197\times 10^{-7}$

$R_{DTR} = R_3 = \frac{\lambda_3}{\mathbf{25760.79034}} = 2.7125\times 10^{-9}$

$R_{GFF} = R_4 = \frac{\lambda_4}{\mathbf{25760.79034}} = 3.1560\times 10^{-8}$

$R_{MaxT} = R_5 = \frac{\lambda_5}{\mathbf{25760.79034}} = 3.7098\times 10^{-8}$

$R_{MinT} = R_6 = \frac{\lambda_6}{25760.79034} = 3.0477\times 10^{-6}$

$R_{PET} = R_7 = \frac{\lambda_7}{25760.79034} = 1.9011\times 10^{-5}$

$R_{RCET} = R_8 = \frac{\lambda_8}{25760.79034} = 3.0969\times 10^{-5}$

$R_{VP} = R_9 = \frac{\lambda_9}{25760.79034} = 6.2165\times 10^{-4}$

$R_{WDF} = R_{10} = \frac{\lambda_{10}}{25760.79034} = 4.9629\times 10^{-3}$

$R_{Precipitation} = R_{11} = \frac{\lambda_{11}}{25760.79034} = 9.9428\times 10^{-1}$

For the data of Amreli

<u>Finding the relative coefficient of variation</u>

Step 1: Finding the covariance matrix:

A covariance matrix is a square matrix giving the covariance between each pair of elements. It is a symmetric matrix with its diagonal entries equal to the variance of the element. Below table is the covariance matrix for the data of Amreli it shows covariance between a pair of parameters. It can be seen from the below table that the value 0.03 shows the covariance of Ground Frost Frequency and Potential Evapotranspiration similarly all the nondiagonal entries shows the covariance between two different parameters . The diagonal entries of the below matrix give the variance of the data of the different parameters. Below matrix is a is a symmetric positive semi definite matrix with non-negative eigen value.

	AT (ºC)	CC (okta)	DTR (ºC)	GFF (Days)	Tmax (ºC)	Tmin (ºC)	PET (mm/unit time)	RCET (mm/unit time)	VP (N/m^2)	WDF (mm/day)	Precipitation (mm)
AT	6.19	28.76	-5.78	-0.39	2.94	8.72	-0.07	0.73	12.80	3.98	110.89
CC	28.76	380.89	-53.04	-2.09	-1.99	51.05	-9.84	-1.42	86.46	55.00	1432.47
DTR	-5.78	-53.04	8.57	0.36	-0.98	-9.55	1.07	-0.19	-15.59	-7.41	-195.54
GFF	-0.39	-2.09	0.36	0.09	-0.19	-0.55	0.03	-0.02	-0.76	-0.27	-7.18
Tmax	2.94	-1.99	-0.98	-0.19	2.51	3.49	0.58	0.67	4.20	-0.59	-12.46
Tmin	8.72	51.05	-9.55	-0.55	3.49	13.04	-0.49	0.85	19.79	6.82	183.11
PET	-0.07	-9.84	1.07	**0.03**	0.58	-0.49	0.45	0.27	-1.04	-1.54	-38.69
RCET	0.73	-1.42	-0.19	-0.02	0.67	0.85	0.27	0.28	1.17	-0.37	-8.09
VP	12.80	86.46	-15.59	-0.76	4.20	19.79	-1.04	1.17	31.45	11.62	314.02
WDF	3.98	55.00	-7.41	-0.27	-0.59	6.82	-1.54	-0.37	11.62	10.37	306.53
PRECIPITATION	110.89	1432.47	-195.54	-7.18	-12.46	183.11	-38.69	-8.09	314.02	306.53	11008.90

Table 7: Covariance Matrix for the data of Amreli

Step 2: Diagonalizing the covariance matrix

22.808	0	0	0	0	0	0	0	0	0	0
0	1.0951	0	0	0	0	0	0	0	0	0
0	0	3.8×10^{-5}	0	0	0	0	0	0	0	0
0	0	0	4.9×10^{-5}	0	0	0	0	0	0	0
0	0	0	0	0.0406	0	0	0	0	0	0
0	0	0	0	0	0.07036	0	0	0	0	0
0	0	0	0	0	0	0.05798	0	0	0	0
0	0	0	0	0	0	0	0.23171	0	0	0
0	0	0	0	0	0	0	0	0.6013	0	0
0	0	0	0	0	0	0	0	0	213.168	0
0	0	0	0	0	0	0	0	0	0	11224.7

Table 8: Diagonalized Covariance Matrix for the data of Amreli

The above Matrix is obtained by diagonalizing the covariance matrix found in step 2. Where the diagonal entries represent the eigen values of the covariance matrix.

Step 3: Finding the relative measure of variation

The relative measure of variation is defined as the ratio of diagonal element to the trace of the diagonal matrix i.e. $R_k = \frac{\lambda_k}{tr(D)}$, 1 ≤k ≤m. The trace of the above diagonal matrix is **11462.75** hence the relative measure of variation is calculated by:

$R_k = \frac{\lambda_k}{11462.75}$. The relative measure of the variation for the above diagonalized matrix is

$R_{AT} = R_1 = \frac{\lambda_1}{11462.75} = 1.99\times 10^{-3}$

$R_{CC} = R_2 = \frac{\lambda_2}{11462.75} = 9.553\times 10^{-5}$

$R_{DTR} = R_3 = \frac{\lambda_3}{11462.75} = 3.3055\times 10^{-9}$

$R_{GFF} = R_4 = \frac{\lambda_4}{11462.75} = 4.3222\times 10^{-8}$

$R_{MaxT} = R_5 = \frac{\lambda_5}{11462.75} = 3.5407\times 10^{-6}$

$R_{MinT} = R_6 = \frac{\lambda_6}{11462.75} = 6.13781\times 10^{-6}$

$R_{PET} = R_7 = \frac{\lambda_7}{11462.75} = 5.05832\times 10^{-6}$

$R_{RCET} = R_8 = \frac{\lambda_8}{11462.75} = 2.0214\times 10^{-5}$

$R_{VP} = R_9 = \frac{\lambda_9}{11462.75} = 5.2458\times 10^{-5}$

$R_{WDF} = R_{10} = \frac{\lambda_{10}}{11462.75} = 1.8596\times 10^{-2}$

$R_{Precipitation} = R_{11} = \frac{\lambda_{11}}{11462.75} = 9.7923\times 10^{-1}$

For the data of Gandhinagar

<u>Finding the relative coefficient of variation</u>

Step 1: Finding the covariance matrix:

A covariance matrix is a square matrix giving the covariance between each pair of elements. It is a symmetric matrix with its diagonal entries equal to the variance of the element. Below table is the covariance matrix for the data of Amreli it shows covariance between a pair of parameters. It can be seen from the below table that the value 1.24 shows the covariance of Minimum Temperature and Potential Evapotranspiration similarly all the nondiagonal entries shows the covariance between two different

parameters. The diagonal entries of the below matrix give the variance of the data of the different parameters. Below matrix is a is a symmetric positive semi definite matrix with non-negative eigen value.

	AT (ºC)	CC (okta)	DTR (ºC)	GFF (Days)	Tmax (ºC)	Tmin (ºC)	PET (mm/unit time)	RCET (mm/unit time)	VP (N/m^2)	WDF (mm/day)	Precipitation (mm)
AT	19.54	47.82	-7.20	-7.20	3.142	23.16	3.14	4.98	26.89	4.59	110.89
CC	47.82	533.96	-77.40	-77.40	-7.89	86.60	-7.89	3.71	159.69	67.39	1867.74
DTR	-7.20	-77.40	12.17	12.17	1.35	-13.29	1.35	-0.30	-24.95	-10.02	-274.24
GFF	-7.20	-77.40	12.17	12.17	1.35	-13.29	1.35	-0.30	-24.95	-10.02	-274.24
Tmax	3.142	-7.89	1.35	1.35	1.24	2.47	**1.24**	1.29	0.56	-1.39	-38.44
Tmin	23.16	86.60	-13.29	-13.29	2.47	29.83	2.47	5.13	39.40	9.62	254.58
PET	3.14	-7.89	1.35	1.35	1.24	2.47	1.24	1.29	-0.56	-1.39	-38.44
RCET	4.98	3.71	-0.30	-0.30	1.29	5.13	1.29	1.64	4.65	-0.06	-2.03
VP	26.89	159.69	-24.95	-24.95	0.56	39.40	0.56	4.65	62.78	11.62	528.13
WDF	4.59	67.39	-10.02	-10.02	-1.39	9.62	-1.39	-0.06	19.39	10.84	319.49
PRECIPITATION	110.89	1867.74	-274.24	-274.24	-38.44	254.58	-38.44	-2.03	528.13	319.49	11072.07

Table 9: Covariance Matrix for the data of Gandhinagar

Step 2: Diagonalizing the covariance matrix

4.3350	0	0	0	0	0	0	0	0	0	0
0	6.6×10^{-16}	0	0	0	0	0	0	0	0	0
0	0	1.42×10^{-4}	0	0	0	0	0	0	0	0
0	0	0	0.00063	0	0	0	0	0	0	0
0	0	0	0	0.0040	0	0	0	0	0	0
0	0	0	0	0	0.09416	0	0	0	0	0
0	0	0	0	0	0	0.95242	0	0	0	0
0	0	0	0	0	0	0	0.65573	0	0	0
0	0	0	0	0	0	0	0	38.954	0	0
0	0	0	0	0	0	0	0	0	261.300	0
0	0	0	0	0	0	0	0	0	0	11451.2

Table 10: Diagonalized Covariance Matrix for the data of Gandhinagar

The above Matrix is obtained by diagonalizing the covariance matrix found in step 2. Where the diagonal entries represent the eigen values of the covariance matrix.

Step 3: Finding the relative measure of variation

The relative measure of variation is defined as the ratio of diagonal element to the trace of the diagonal matrix i.e. $R_k = \frac{\lambda_k}{tr(D)}$, 1≤k≤m. The trace of the above diagonal matrix is 11757.48 hence the relative measure of variation is calculated by:

$R_k = \frac{\lambda_k}{11757.48}$. The relative measure of the variation for the above diagonalized matrix is

$R_{AT} = R_1 = \frac{\lambda_1}{11757.48} = 3.6873 \times 10^{-4}$

$R_{CC} = R_2 = \frac{\lambda_2}{11757.48} = 5.6656 \times 10^{-20}$

$R_{DTR} = R_3 = \frac{\lambda_3}{11757.48} = 1.2058 \times 10^{-8}$

$R_{GFF} = R_4 = \frac{\lambda_4}{11757.48} = 5.3478 \times 10^{-8}$

$R_{MaxT} = R_5 = \frac{\lambda_5}{11757.48} = 3.4193 \times 10^{-7}$

$R_{MinT} = R_6 = \frac{\lambda_6}{11757.48} = 8.0076 \times 10^{-6}$

$R_{PET} = R_7 = \frac{\lambda_7}{11757.48} = 8.1006 \times 10^{-5}$

$R_{RCET} = R_8 = \frac{\lambda_8}{11757.48} = 5.5771 \times 10^{-5}$

$R_{VP} = R_9 = \frac{\lambda_9}{11757.48} = 3.3131 \times 10^{-3}$

$R_{WDF} = R_{10} = \frac{\lambda_{10}}{11757.48} = 2.2224 \times 10^{-2}$

$R_{Precipitation} = R_{11} = \frac{\lambda_{11}}{11757.48} = 9.7394 \times 10^{-1}$

Further conclusions can be made from the above methodology.

CHAPTER 3: RESULTS AND DISCUSSION

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3.4 CONCLUSION: _____52

This chapter includes the results obtained by applying the research methodology to the data of Surat, Amreli and Gandhinagar followed by the discussion of the results and a conclusion.

3.1 RESULTS AND DISCUSSION FOR THE DATA OF SURAT

Correlation Matrix for the data of Surat

	AT	CC	DTR	GFF	Tmax	Tmin	PET	RCET	VP	WDF	Precipitation
AT	1	0.382	-0.523	-0.5	0.785	0.928	0.417	0.791	0.656	0.247	0.212
CC	0.382	1	-0.963	-0.398	-0.255	0.687	-0.6	-0.092	0.877	0.929	0.815
DTR	-0.523	-0.963	1	0.424	0.118	-0.802	0.511	-0.004	-0.926	-0.882	-0.767
GFF	-0.5	-0.398	0.424	1	-0.273	-0.535	0.101	-0.167	-0.474	-0.339	-0.287
Tmax	0.785	-0.255	0.118	-0.273	1	0.499	0.856	0.918	0.092	-0.353	-0.31
Tmin	0.928	0.687	-0.802	-0.535	0.499	1	0.07	0.556	0.864	0.558	0.483
PRT	0.417	-0.6	0.511	0.101	0.856	0.07	1	0.849	-0.279	-0.643	-0.546
RCET	0.791	-0.092	-0.004	-0.167	0.918	0.556	0.849	1	0.234	-0.202	-0.164
VP	**0.656**	**0.877**	-0.926	-0.474	0.092	**0.864**	-0.279	0.234	1	**0.781**	0.686
WDF	0.247	**0.929**	-0.882	-0.339	-0.353	0.558	-0.643	-0.202	**0.781**	1	0.94
PRECIPITATION	0.212	0.815	-0.767	-0.287	-0.31	0.483	-0.546	-0.164	0.686	0.94	1

It is evident from the above Correlation matrix that Wet Day Frequency is strongly correlated with Cloud Cover and Vapor Pressure as correlation coefficient of Wet Day Frequency with Cloud Cover and Vapor Pressure is 0.929 and 0.781 respectively. whereas, Vapor Pressure is related with Cloud Cover, Minimum Temperature, Wet Day Frequency and Average Temperature as correlation coefficient of Vapor Pressure with Cloud Cover, Minimum Temperature, Wet Day Frequency and Average Temperature is 0.877,0.864,0.781,0.656 respectively. Hence it can be said that Wet Day Frequency depends on Cloud Cover while Vapor Pressure depends on Cover, Minimum Temperature, Wet Day Frequency and Average Temperature.

Coefficient of variation for the data of Surat

R_{AT}	R_{CC}	R_{DTR}	R_{GFF}	R_{MaxT}	R_{MinT}	R_{PET}	R_{RCET}	R_{VP}	R_{WDF}	$R_{Precipitation}$
8.1028×10^{-5}	1.0197×10^{-7}	2.7125×10^{-8}	3.1560×10^{-8}	3.7098×10^{-8}	3.0477×10^{-6}	1.9011×10^{-5}	3.0969×10^{-5}	6.2165×10^{-4}	4.9629×10^{-3}	9.9428×10^{-1}

Table 11: Coefficient of variation for the data of Surat

The reduced dimension of the matrix can be known by using the relative coefficient of variation. If the relative coefficient of variation is less than some tolerance value ε than that corresponding column and row can be deleted from the original matrix resulting into a matrix of reduced dimension.

Let us consider the tolerance level as $\varepsilon = \mathbf{10^{-4}}$ hence the column and row corresponding to the value of relative measure of variation less that can be deleted. Initially we had a 11×11 matrix which can be reduced to a 3×3 matrix as the value of $\boldsymbol{R_{VP}}, \boldsymbol{R_{WDF}}, \boldsymbol{R_{Precipitation}}$ is greater than ε, while the value of relative coefficient of variation of other parameters is less than ε.

3.2 RESULTS AND DISCUSSION FOR THE DATA OF AMRELI

Correlation Matrix for the data of Amreli

	AT	CC	DTR	GFF	Tmax	Tmin	PET	RCET	VP	WDF	Precipitation
AT	1	0.592	-0.794	-0.524	0.745	**0.97**	-0.04	0.556	**0.917**	0.497	0.425
CC	0.592	1	-0.928	-0.359	-0.064	0.724	-0.748	-0.138	0.79	0.875	0.7
DTR	-0.794	-0.928	1	0.408	-0.211	-0.904	0.544	-0.121	-0.949	-0.786	-0.636
GFF	-0.524	-0.359	0.408	1	-0.409	-0.511	0.17	-0.146	-0.451	-0.284	-0.229
Tmax	0.745	-0.064	0.211	-0.409	1	0.609	0.545	0.793	0.473	-0.115	-0.075
Tmin	0.97	0.724	-0.904	-0.511	0.609	1	-0.202	0.446	0.977	0.587	0.483
PRT	-0.04	-0.748	0.544	0.17	0.545	-0.202	1	0.756	-0.275	-0.71	-0.547
RCET	0.556	-0.138	-0.121	-0.146	0.793	0.446	0.756	1	0.394	-0.215	-0.146
VP	0.917	0.79	-0.949	-0.451	0.473	0.977	-0.275	0.394	1	0.643	0.534
WDF	0.497	**0.875**	-0.786	-0.284	-0.115	0.587	-0.71	-0.215	**0.643**	1	0.907
PRECIPITATION	0.425	0.7	-0.636	-0.229	-0.075	0.483	-0.547	-0.146	0.534	0.907	1

From the above correlation matrix, it can be clearly seen that for Amreli Wet Day Frequency is strongly correlated with Cloud Cover and Vapor Pressure as correlation coefficient of Wet Day Frequency with Cloud Cover and Vapor Pressure is 0.875 and 0.643 respectively. while Average Temperature is Strongly correlated with Minimum Temperature and Vapor Pressure as correlation coefficient of Average Temperature with Minimum Temperature and Vapor

Pressure is 0.97 and 0.917 . Hence, it can be said that Wet Day Frequency depends on Cloud Cover and Average Temperature depends on Minimum Temperature and Vapor Pressure.

Coefficient of variation for the data of Amreli

R_{AT}	R_{CC}	R_{DTR}	R_{GFF}	R_{MaxT}	R_{MinT}	R_{PET}	R_{RCET}	R_{VP}	R_{WDF}	$R_{Precipitation}$
1.9897×10^{-3}	9.55346×10^{-5}	3.3055×10^{-9}	4.3221×10^{-8}	3.5407×10^{-6}	6.1378×10^{-6}	5.0583×10^{-6}	2.0214×10^{-5}	5.246×10^{-5}	1.8597×10^{-2}	9.7923×10^{-1}

Table 12: Coefficient of variation for the data of Amreli

Let us consider the tolerance level as $\varepsilon = \mathbf{10^{-4}}$ hence the column and row corresponding to the value of relative measure of variation less that can be deleted. Initially we had a 11×11 matrix which can be reduced to a 3×3 matrix as the value of $\boldsymbol{R_{AT}}, \boldsymbol{R_{WDF}}, \boldsymbol{R_{Precipitation}}$ is greater than ε, while the value of relative coefficient of variation of other parameters is less than ε.

3.3 RESULTS AND DISCUSSION FOR THE DATA OF GANDHINAGAR

Correlation Matrix for the data of Gandhinagar

	AT	CC	DTR	GFF	Tmax	Tmin	PET	RCET	VP	WDF	Precipitation
AT	1	0.47	-0.47	-0.47	0.64	**0.96**	0.64	0.88	**0.77**	0.32	0.25
CC	0.47	1	-0.96	-0.96	-0.31	0.69	-0.31	0.13	0.87	0.89	0.77
DTR	-0.47	-0.96	1	1.00	0.35	-0.70	0.35	-0.07	-0.90	-0.87	-0.75
GFF	-0.47	-0.96	1.00	1	0.35	-0.70	0.35	-0.07	-0.90	-0.87	-0.75
Tmax	0.64	-0.31	0.35	0.35	1	0.41	1.00	0.90	0.06	-0.38	-0.33
Tmin	0.96	0.69	-0.70	-0.70	0.41	1	0.41	0.73	0.91	0.53	0.44
PET	0.64	-0.31	0.35	0.35	1.00	0.41	1	0.90	0.06	-0.38	-0.33
RCET	0.88	0.13	-0.07	-0.07	0.90	0.73	0.90	1	0.46	-0.01	-0.02
VP	**0.77**	**0.87**	-0.90	-0.90	0.06	**0.91**	0.06	0.46	1	**0.74**	0.63
WDF	0.32	**0.89**	-0.87	-0.87	-0.38	0.53	-0.38	-0.01	**0.74**	1	0.92
PRECIPITATION	0.25	0.77	-0.75	-0.75	-0.33	0.44	-0.33	-0.02	0.63	0.92	1

It is evident from the above correlation matrix that Wet Day Frequency is strongly correlated with Cloud Cover and Vapor Pressure as the correlation coefficient of Wet Day Frequency with Cloud Cover and Vapor Pressure is 0.89 and 0.74 respectively. Average Temperature is strongly correlated with Minimum Temperature and Vapor Pressure as correlation coefficient of Average temperature with Minimum Temperature and Vapor Pressure is 0.96 and 0.77, Vapor Pressure is related with Minimum Temperature , Cloud Cover, Average Temperature and Wet Day Frequency as correlation coefficient of Vapor Pressure with Minimum Temperature , Cloud Cover, Average Temperature and Wet Day Frequency is 0.91,0.87,0.77,0.74 respectively. Hence it can be said that Wet Day Frequency Depends on Cloud Cover , Average Temperature Depends on Minimum Temperature and Vapor Pressure and Vapor Pressure depends on Minimum Temperature , Cloud Cover, Average Temperature and Wet Day Frequency.

Coefficient of variation for the data of Gandhinagar

R_{AT}	R_{CC}	R_{DTR}	R_{GFF}	R_{MaxT}	R_{MinT}	R_{PET}	R_{RCET}	R_{VP}	R_{WDF}	$R_{Precipitation}$
3.6873×10^{-4}	5.6656×10^{-20}	1.2058×10^{-8}	5.3478×10^{-8}	3.4193×10^{-7}	8.0076×10^{-6}	8.1006×10^{-5}	5.5771×10^{-5}	3.3131×10^{-3}	2.2224×10^{-2}	9.7394×10^{-1}

Table 13: Coefficient of variation for the data of Gandhinagar

Let us consider the tolerance level as $\varepsilon = \mathbf{10^{-4}}$ hence the column and row corresponding to the value of relative measure of variation less that can be deleted. Initially we had a 11×11 matrix which can be reduced to a 4×4 matrix as the value of $\boldsymbol{R_{AT}}$, $\boldsymbol{R_{WDF}}$, $\boldsymbol{R_{VP}}$, $\boldsymbol{R_{Precipitation}}$ is greater than ε, while the value of relative coefficient of variation of other parameters is less than ε.

3.4 CONCLUSION

From the correlation matrix for the data of Surat, Amreli and Gandhinagar it can be concluded that Wet Day Frequency depends on Cloud Cover and Vapor Pressure, Vapor Pressure depends on Average Temperature, Cloud Cover, Minimum Temperature and Wet Day Frequency and Average Temperature depends on Minimum Temperature and Vapor Pressure.

As discussed in 3.1,3.2,3.3 Precipitation depends on Wet Day Frequency and Vapor Pressure for the Data of Surat, Precipitation depends on Wet Day Frequency and Average Temperature for the Data of Amreli, Precipitation depends on Wet Day Frequency, Vapor Pressure and Average Temperature for the Data of Gandhinagar respectively. Hence Wet Day Frequency and Vapor Pressure acts as the predominant parameters for the data of Surat, Wet Day Frequency and Average Temperature acts as the predominant parameters for the data of Amreli, and Wet Day Frequency, Vapor Pressure and Average Temperature acts as the predominant parameters for the data of Gandhinagar. The variation in predominant parameters is due to the different geographical conditions of all the three areas. Surat is a coastal region and Gandhinagar lies on the bank of river hence, Vapor Pressure plays an important role in influencing Precipitation over there, while Amreli is a plain land hence, Vapor Pressure does not play a crucial role in influencing rainfall.

This study can be useful in rainfall prediction whenever the data of all the parameters influencing the rainfall is not available, the rainfall can be predicted by using the availability of the data of the predominant parameters. This can also be useful for Meteorological department, weather forecasting, Flood management, Drought management and in the agricultural area.

APPENDIX A

A Python code to find the Correlation Matrix from the given data

```
import matplotlib.pyplot as plt
import pandas as pd
a=pd.read_excel("/merged data.xlsx")
df=pd.DataFrame(a)
corrMatrix=a.corr()
print("correlation Matrix= ",corrMatrix)
```

APPENDIX B

A Python code to find the Covariance Matrix from the given data, followed by its diagonalization and finding the relative measure of variation for the diagonalized Covariance Matrix.

```
import pandas as pd
import numpy as np
from sympy import *
a=pd.read_excel("/content/mergedsurat data.xlsx")
df=pd.DataFrame(a)
cm=df.cov()
print("covariance matrix=",cm.round(3))
C=Matrix(cm)
p,d=C.diagonalize()
print("Diagonalized matrix : {}".format(d))
D=np.diag(d)
print("Diagonal elements=",D)
t=np.trace(d)
print("Trace=",t)
R=list(map(lambda D:D/t,D))
print("R=",R)
```

BIBLIOGRAPHY

1. Atlles, F. J., Alonso and J., L´opez, G. (2014). Cloud cover forecasting from METEOSAT data. *Energy Procedia*, **57**, 1317-1320.

2. Chen, Fan, and Peter J. Robinson. (2009) Estimating reference crop evapotranspiration with ETgages. *Journal of irrigation and drainage engineering* **135**, no. 3: 335-342.

3. Espinoza, et al. (2016) Evolution of wet-day and dry-day frequency in the western Amazon basin: Relationship with atmospheric circulation and impacts on vegetation. *Water Resources Research* **52.11**: 8546-8560.

4. Eyra AE, Melody AD and Hakii N. (2015) Precipitation and Rainfall Types with Their Characteristic, *Journal of Natural Sciences Research*, **5**, 89–91.

5. Hejazizadeh, Z. and Naserzadeh, M. H. (2007). Calculation and analysis of frost duration times by using Delphi programming: a case study in Lorestan, Iran. *Journal of Applied Sciences,* **7**(23), 3831-3835.

6. Katavoutas, G., Founda, D., Varotsos, K.V. and Giannakopoulos C., (2023). Diurnal Temperature Range and Its Response to Heat Waves in 16 European Cities—Current and Future Trends. *Sustainability*, **15**(17), p.12715.

7. Narayan, S. and Mittal, P.K., (2010). *A textbook of matrices*, S. Chand Publishing.

8. Shao, J., (2003). *Mathematical statistics*, Springer Science & Business Media.

9. Stoll, R.R., (2013). *Linear algebra and matrix theory*, Courier Corporation.